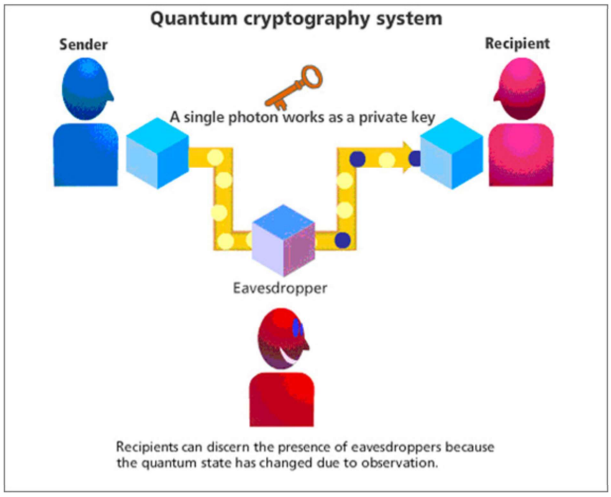
Quantum Cryptography



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Covered in detail in the previous note as they relate to CISSP Domain 3o

Crypto history



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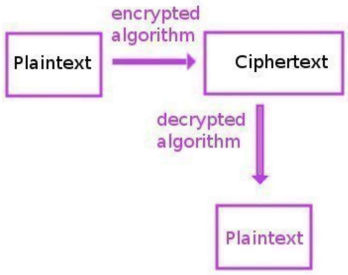
Reference : https://en.wikipedia.org/wiki/History_of_cryptography

Key Encryption Concepts and Definitions

Plaintext or clear-text **“My name is Abu Bakr”**

Encrypted text also known as Ciphertext

ACNHFUK”{}”safsdfsdg1236547856

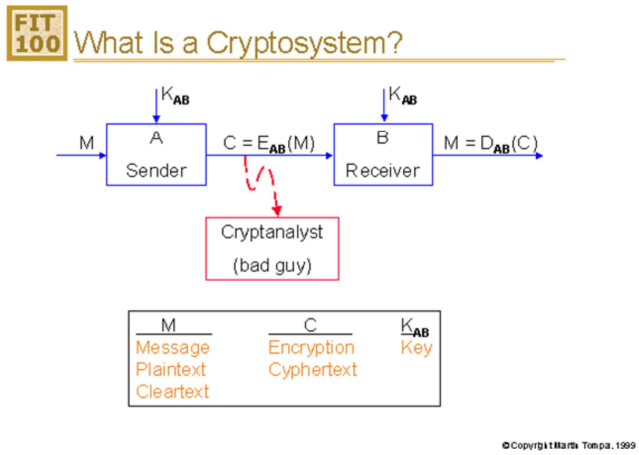


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Plaintext or cleartext: This is the message or data in its natural format and in readable form. Plaintext is human readable and is extremely vulnerable from a confidentiality perspective. Plaintext is the message or data that has not been turned into a secret.

Ciphertext or cryptogram: This is the altered form of a plaintext message so as to be unreadable for anyone except the intended recipients. In other words, it has been turned into a secret. An attacker seeing ciphertext would be unable to easily read the message or to determine its content. Also referred to as the message that has been turned into a secret.

Cryptosystem

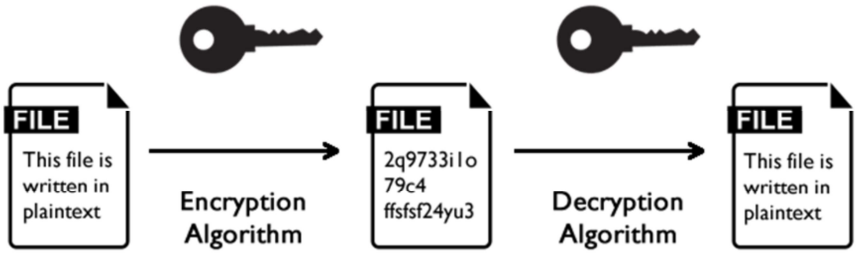


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Cryptosystem: This represents the entire cryptographic operation and system. This typically includes the algorithm, key, and key management functions, together with the services that can be provided through cryptography. The cryptosystem is the complete set of applications that allows sender and receiver to communicate using cryptography systems.

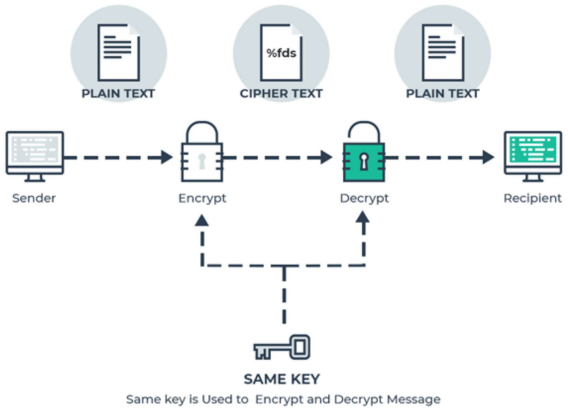
Crypto Algorithm



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Algorithm: An algorithm is a mathematical function that is used in the encryption and decryption processes. It may be quite simple or extremely complex. Also defined as the set of instructions by which encryption and decryption is done.

Encryption and Decryption



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Encryption: This is the process and act of converting the message from its plaintext to ciphertext. Sometimes this is also referred to as enciphering. The two terms are sometimes used interchangeably in the literature and have similar meanings.

Decryption: This is the reverse process from encryption. It is the process of converting a ciphertext message back into
plaintext through the use of the cryptographic algorithm and key (cryptovariable) that was used to do the original
encryption. This term is also used interchangeably with the term deciphering.

Key or crypto variable

Crypto variable - NSA calls the output of a stream cipher a **key** or **key** stream. It often uses the term **crypto variable** for the bits that control the stream cipher, what the public cryptographic community calls a **key**. data encryption **key** (DEK) used to encrypt the underlying data. And to decrypt a message data decryption key is used (DDK).

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Key or cryptovariable: The input that controls the operation of the cryptographic algorithm. It determines the behavior of the algorithm and permits the reliable encryption and decryption of the message. There are both secret and public keys used in cryptographic algorithms.